

Photons, Atoms and Nuclei

Where classical physics breaks down: photons, matter waves, the quantised atom, the nucleus, and radiation in medicine.

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By 1900, classical physics seemed almost complete. A handful of puzzles remained: the spectrum of hot objects, the way light ejects electrons from metals, and the sharp coloured lines emitted by gases. Resolving them overturned our picture of nature. Light turned out to come in packets (**photons**), matter turned out to behave like waves, atoms have **quantised** energy levels, and the tiny atomic nucleus stores energies a million times larger than chemical bonds. This final chapter introduces these ideas and their consequences: X-ray imaging and CT, electron microscopy, lasers and spectroscopy, nuclear power, radiometric dating, and nuclear medicine, including PET scans and radiotherapy.

YOU WILL BE ABLE TO

- Use photon energy and momentum, $E = hf = hc/\lambda$ and $p = h/\lambda$.
- Analyse the photoelectric effect with Einstein's equation.
- Describe X-ray production, Compton scattering and exponential attenuation, as used in radiography and CT.
- Use the de Broglie wavelength and the uncertainty principle.
- Apply the Bohr model of hydrogen: energy levels, spectra and ionisation.
- Describe nuclear structure, binding energy, fission and fusion, using $E = mc^2$.
- Apply the radioactive decay law, half-life and activity, to dating and nuclear medicine, and use dose units for radiation safety.

THINK FIRST

Ultraviolet light gives you sunburn, but no amount of visible red light will, even very intense red light. A sunbed with a weak UV lamp can burn you in minutes, while sitting in front of a powerful red heat lamp cannot. Why does the *colour* matter more than the *brightness*?

Light is absorbed in **photons**, each with energy $E = hf$. Damaging a DNA molecule needs a few electron-volts delivered to *one* molecule at once. A UV-B photon (about 300 nm) carries about 4 eV, enough to break the bond on its own. A red photon (about 700 nm) carries only about 1.8 eV. A brighter red light delivers *more* photons, but each one is still too weak, and the molecule cannot save up energy from several photons. Intensity sets *how many* photons arrive; frequency sets *what each one can do*. The same idea explains why X-rays and gamma rays ionise and damage tissue, while radio waves of any intensity only warm it.

§24.1

Photons

Max Planck (1900) explained the spectrum of thermal radiation by assuming that energy is exchanged in discrete amounts. Albert Einstein (1905) went further: light itself consists of **photons**, each carrying

$$E = hf = \frac{hc}{\lambda}, \quad p = \frac{E}{c} = \frac{h}{\lambda},$$

where $h = 6.626 \times 10^{-34}$ J s is **Planck's constant**. A useful shortcut is $hc = 1240$ eV nm, so a photon of wavelength λ (in nm) has energy $E = 1240/\lambda$ eV.

RADIATION	WAVELENGTH	PHOTON ENERGY
FM radio	3 m	4×10^{-7} eV
Red light	650 nm	1.9 eV
Violet light	400 nm	3.1 eV
UV-B	300 nm	4.1 eV
Diagnostic X-ray	0.012–0.1 nm	10–100 keV

RADIATION	WAVELENGTH	PHOTON ENERGY
PET annihilation photon	2.4 pm	511 keV
Radiotherapy beam	below 1 pm	1–20 MeV

Photons with energies above about 10 eV can ionise atoms and break chemical bonds. This is the boundary between **non-ionising** and **ionising** radiation.

§24.2

The photoelectric effect

Light shining on a metal can eject electrons. Experiments showed three facts that the wave theory of light could not explain:

1. **There is a threshold frequency.** Below it, no electrons are emitted, however intense the light.
2. **The maximum kinetic energy of the electrons depends on the frequency,** not on the intensity. The intensity changes only the *number* of electrons.
3. **Emission is instantaneous,** even in very dim light.

Einstein's explanation: each photon gives *all* its energy to one electron. The electron must spend at least the **work function** ϕ to escape the metal, so

$$K_{\max} = hf - \phi.$$

The threshold frequency is $f_0 = \phi/h$. The maximum kinetic energy is measured by the **stopping potential** V_s , the reverse voltage that just stops the most energetic electrons: $eV_s = K_{\max}$.

WORKED EXAMPLE 24.1

Light on sodium

Sodium has a work function of 2.28 eV. Violet light of 400 nm falls on it. Find the maximum kinetic energy of the photoelectrons, the stopping potential, and the threshold wavelength.

Solution

$$E = \frac{1240}{400} = 3.10 \text{ eV}, \quad K_{\max} = 3.10 - 2.28 = 0.82 \text{ eV}, \quad V_s = 0.82 \text{ V}.$$

$$\lambda_0 = \frac{1240}{2.28} = 544 \text{ nm}.$$

Light of longer wavelength (yellow, red) ejects no electrons, however bright it is.

IN PRACTICE

Detecting photons. Photomultiplier tubes use the photoelectric effect and then multiply each ejected electron about a million-fold. They sit behind the scintillation crystals of gamma cameras and PET scanners, turning individual 140 keV or 511 keV photons into measurable pulses. Digital cameras, X-ray flat-panel detectors and pulse oximeters use the closely related *internal* photoelectric effect in semiconductors. Solar cells use it to generate electricity.

§24.3

X-rays

Production. In an X-ray tube, electrons accelerated through a voltage V strike a metal target, usually tungsten. Two processes produce X-rays:

- **Bremsstrahlung** (“braking radiation”). Electrons decelerating in the fields of the nuclei produce a continuous spectrum, up to a maximum photon energy eV . This gives the Duane–Hunt limit $\lambda_{\min} = hc/eV$ (Chapter 15).
- **Characteristic lines.** Incoming electrons knock out inner-shell electrons, and outer electrons dropping into the vacancies emit X-rays at sharp energies that are characteristic of the element.

Compton scattering. When an X-ray photon scatters off a loosely bound electron, it gives up some of its energy and momentum like a billiard ball. Conserving energy and momentum (with relativistic mechanics) gives the wavelength shift

$$\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta), \quad \frac{h}{m_e c} = 2.43 \text{ pm}.$$

Arthur Compton’s 1923 measurement of this shift was decisive evidence that photons carry momentum.

Attenuation. A narrow beam of X-rays passing through matter loses intensity exponentially, through photoelectric absorption and Compton scattering:

$$I = I_0 e^{-\mu x}.$$

The **attenuation coefficient** μ depends on the material and the photon energy. The **half-value layer**, $x_{1/2} = \ln 2 / \mu$, is the thickness that halves the intensity. Bone, rich in calcium, attenuates much more than soft tissue. Radiographs are maps of this contrast, and **computed tomography** (CT) reconstructs μ in three dimensions from hundreds of projections.

WORKED EXAMPLE 24.2

Compton scattering of a diagnostic X-ray

A 100 keV X-ray photon scatters through 90° from an electron in tissue. Find the energy of the scattered photon and the kinetic energy given to the electron.

Solution

$$\lambda = \frac{1240 \text{ eV nm}}{1.0 \times 10^5 \text{ eV}} = 12.4 \text{ pm}, \quad \lambda' = 12.4 + 2.43(1 - 0) = 14.8 \text{ pm}.$$

$$E' = \frac{1240 \text{ eV nm}}{14.8 \times 10^{-3} \text{ nm}} = 84 \text{ keV}, \quad K_e = 100 - 84 = 16 \text{ keV}.$$

Evaluate. Compton-scattered photons fog radiographic images, and they are the main source of radiation dose to staff standing beside the patient. *Anti-scatter grids* in front of the detector block photons that arrive at an angle, and staff wear lead aprons.

WORKED EXAMPLE 24.3

X-ray contrast

At 60 keV, take $\mu \approx 0.20 \text{ cm}^{-1}$ for soft tissue and $\mu \approx 0.60 \text{ cm}^{-1}$ for bone. Find the fraction transmitted through 20 cm of soft tissue, and through 18 cm of soft tissue plus 2 cm of bone. Find the half-value layer of soft tissue.

Solution

$$T_{\text{tissue}} = e^{-0.20(20)} = e^{-4.0} = 1.8\%, \quad T_{\text{with bone}} = e^{-[0.20(18)+0.60(2)]} = e^{-4.8} = 0.82\%.$$

$$x_{1/2} = \frac{\ln 2}{0.20} = 3.5 \text{ cm}.$$

Evaluate. Two centimetres of bone halve the transmitted beam, which shows up clearly as a lighter “shadow” on the image. Only about 1–2% of the beam gets through the body at all. The rest is absorbed, which is what delivers the radiation dose.

§24.4

Matter waves

Louis de Broglie (1924) proposed that, if light waves behave like particles, then particles should behave like waves, with wavelength

$$\lambda = \frac{h}{p}.$$

Davisson and Germer confirmed this in 1927, when they diffracted electrons from a nickel crystal. Electrons, neutrons, atoms and even large molecules all show interference.

For macroscopic objects, λ is absurdly small. A thrown baseball has $\lambda \sim 10^{-34}$ m, so its wave nature is completely unobservable.

WORKED EXAMPLE 24.4

Wavelengths of electrons

(a) Find the de Broglie wavelength of an electron accelerated through 54 V (the Davisson–Germer experiment). (b) An electron microscope accelerates electrons through 200 kV. Find their wavelength, taking relativity into account, using $pc = \sqrt{K^2 + 2Km_e c^2}$ with $m_e c^2 = 511$ keV.

Solution

(a) Non-relativistic, since $K = 54$ eV:

$$p = \sqrt{2m_e K} = \sqrt{2(9.11 \times 10^{-31})(54 \times 1.602 \times 10^{-19})} = 3.97 \times 10^{-24} \text{ kg m/s}, \quad \lambda = \frac{h}{p}$$

which is comparable to atomic spacings, so crystals diffract the electrons.

(b) Relativistic:

$$pc = \sqrt{200^2 + 2(200)(511)} = \sqrt{244\,400} = 494 \text{ keV}, \quad \lambda = \frac{hc}{pc} = \frac{1240 \text{ eV nm}}{494\,000 \text{ eV}} = 2.5 \times 10^{-4} \text{ nm}$$

Evaluate. That wavelength is 200 000 times shorter than visible light. Electron microscopes are limited mainly by lens aberrations, not by diffraction, and they still reach atomic resolution. **Cryo-electron microscopy** (Nobel Prize, 2017) images frozen proteins and viruses at near-atomic detail, which is how the structure of the SARS-CoV-2 spike protein was determined within weeks in 2020.

The uncertainty principle

A wave confined to a region of size Δx must contain a spread of wavelengths, and so a spread of momenta. Werner Heisenberg showed that

$$\Delta x \Delta p_x \geq \frac{\hbar}{2}, \quad \hbar = \frac{h}{2\pi} = 1.055 \times 10^{-34} \text{ J s.}$$

This is not a limitation of our instruments. It is a property of nature. Confining a particle in a smaller region forces it to have a larger momentum, and so a larger kinetic energy. This “zero-point” energy explains why atoms are stable: an electron squeezed into the nucleus would need an enormous kinetic energy. It also explains why atomic energies are electron-volts while nuclear energies are mega-electron-volts (Problem P24.2 and Example 24.5).

WORKED EXAMPLE 24.5

Why atomic energies are electron-volts

Estimate the minimum kinetic energy of an electron confined to a region of 0.10 nm (the size of an atom), and of a proton confined to 5.0 fm (the size of a nucleus).

Solution

Electron:

$$\Delta p \approx \frac{\hbar}{2\Delta x} = \frac{1.055 \times 10^{-34}}{2(1.0 \times 10^{-10})} = 5.3 \times 10^{-25} \text{ kg m/s,}$$

$$K \approx \frac{(\Delta p)^2}{2m_e} = 1.5 \times 10^{-19} \text{ J} \approx 1 \text{ eV.}$$

Proton:

$$\Delta p \approx \frac{1.055 \times 10^{-34}}{1.0 \times 10^{-14}} = 1.1 \times 10^{-20} \text{ kg m/s,} \quad K \approx \frac{(\Delta p)^2}{2m_p} = 3.3 \times 10^{-14} \text{ J} \approx 0.2 \text{ MeV.}$$

(A fuller treatment, with larger momentum spreads, gives a few MeV.)

Evaluate. These rough estimates correctly predict that chemistry involves energies of electron-volts and nuclear physics involves mega-electron-volts. That million-fold ratio is why nuclear fuel is a million times more energetic than chemical fuel.

§24.5

The Bohr model of hydrogen

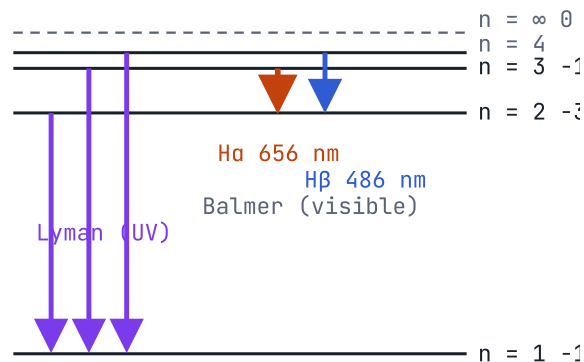
Heated hydrogen gas emits light only at certain sharp wavelengths, the most famous being the red H- α line at 656 nm. Niels Bohr (1913) explained this by postulating that the electron can occupy only orbits whose angular momentum is a whole number of units of $\hbar$: $L = n\hbar$. Combining this with Coulomb's law (Problem P24.14) gives the allowed radii and energies:

$$r_n = n^2 a_0, \quad a_0 = 0.0529 \text{ nm}; \quad E_n = -\frac{13.6 \text{ eV}}{n^2}, \quad n = 1, 2, 3, \dots$$

When the electron drops from level n_i to a lower level n_f , it emits a photon carrying the energy difference:

$$hf = E_{n_i} - E_{n_f} = 13.6 \text{ eV} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right).$$

Transitions ending on $n_f = 1$ form the Lyman series (ultraviolet). Those ending on $n_f = 2$ form the Balmer series (visible). The **ionisation energy**, the energy needed to remove the electron from the ground state, is 13.6 eV.



Energy levels of hydrogen, not to scale near the top. Downward jumps emit photons carrying the energy difference: the Lyman series (to $n = 1$) is ultraviolet, and the Balmer series (to $n = 2$) is visible.

WORKED EXAMPLE 24.6

The red line of hydrogen

Find the wavelength of the photon emitted when a hydrogen atom drops from $n = 3$ to $n = 2$.

Solution

$$E = 13.6 \left(\frac{1}{4} - \frac{1}{9} \right) = 1.89 \text{ eV}, \quad \lambda = \frac{1240}{1.89} = 656 \text{ nm (H-, red)}.$$

The Bohr model works exactly only for hydrogen-like atoms, and it was superseded by quantum mechanics (Schrödinger, 1926). Its central idea, **discrete energy levels** with photons carrying the differences, remains the basis of all spectroscopy. It explains why each element has a unique spectral “fingerprint”, used to identify elements in stars, in blood samples and in forensic traces. **Lasers** are based on *stimulated emission*: a photon of the right energy triggers an excited atom to emit an identical photon, in phase and travelling in the same direction, producing intense, coherent light. Lasers are used in surgery, ophthalmology, dermatology and dentistry.

§24.6

The nucleus

A nucleus contains Z protons and N neutrons, with mass number $A = Z + N$. It is written A_ZX , for example ${}^{12}_6\text{C}$. **Isotopes** of an element have the same Z but different N . Nuclei are tiny, with radius

$$R \approx (1.2 \text{ fm}) A^{1/3},$$

so all nuclei have about the same density, about $2 \times 10^{17} \text{ kg/m}^3$. The protons repel each other enormously, and they are held together by the short-range **strong nuclear force**.

Binding energy and $E = mc^2$

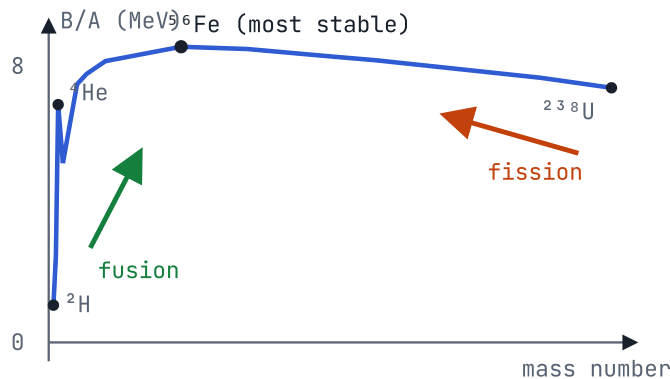
A nucleus has *less* mass than the protons and neutrons it is made of. The **mass defect** Δm corresponds to the **binding energy**, the energy released in assembling the nucleus, or needed to pull it apart:

$$E_B = \Delta m c^2, \quad 1 \text{ u} = 931.5 \text{ MeV}/c^2.$$

The **binding energy per nucleon**, E_B/A , rises steeply for light nuclei, peaks near iron ($A \approx 56$, about 8.8 MeV), and falls gently for heavy nuclei. This curve explains both kinds of nuclear energy:

- **Fission**: splitting a heavy nucleus such as uranium into two medium nuclei moves the nucleons up the curve, releasing about 200 MeV per fission.

- **Fusion:** joining light nuclei, such as the hydrogen isotopes, into helium releases even more energy *per kilogram* of fuel. Fusion powers the Sun.



Binding energy per nucleon. Iron-56 sits near the peak. Fusing light nuclei, or splitting heavy ones, moves towards the peak and releases energy.

WORKED EXAMPLE 24.7

The binding energy of helium-4

Using atomic masses, $m(^1\text{H}) = 1.007825 \text{ u}$, $m_n = 1.008665 \text{ u}$ and $m(^4\text{He}) = 4.002603 \text{ u}$, find the binding energy of ^4He and its binding energy per nucleon.

Solution

Using atomic masses (which include the electrons) for both H and He makes the electron masses cancel.

$$\Delta m = 2(1.007825) + 2(1.008665) - 4.002603 = 4.032980 - 4.002603 = 0.030377 \text{ u.}$$

$$E_B = 0.030377 \times 931.5 = 28.3 \text{ MeV}, \quad \frac{E_B}{A} = 7.07 \text{ MeV.}$$

Evaluate. The alpha particle is exceptionally tightly bound for such a light nucleus. That is why alpha particles are emitted as a single unit in radioactive decay, and why helium is the main product of fusion in stars.

§24.7

Radioactivity

Unstable nuclei decay spontaneously, emitting radiation:

DECAY	EMITTED	CHANGE IN NUCLEUS	PENETRATION
Alpha (α)	${}^4\text{He}$ nucleus	$Z - 2, A - 4$	Stopped by paper or the outer skin. Very damaging if ingested.
Beta-minus (β^-)	electron + antineutrino	$Z + 1$ (a neutron becomes a proton)	A few mm of tissue
Beta-plus (β^+)	positron + neutrino	$Z - 1$ (a proton becomes a neutron)	Annihilates within about 1 mm, giving two 511 keV photons
Gamma (γ)	high-energy photon	none; the nucleus drops to a lower energy level	Very penetrating. Shielded by lead or concrete.

The decay law

Each nucleus has a constant probability per unit time, λ , of decaying, independent of its history. So the rate of decay is proportional to the number of nuclei present:

$$\frac{dN}{dt} = -\lambda N \implies N = N_0 e^{-\lambda t}.$$

The **half-life** is the time for half the nuclei to decay:

$$t_{1/2} = \frac{\ln 2}{\lambda}.$$

The **activity** $A = \lambda N$ is the number of decays per second. It is measured in **becquerels** (1 Bq = 1 decay/s). The older unit is the curie: 1 Ci = 3.7×10^{10} Bq. Activity decays with the same half-life: $A = A_0 \left(\frac{1}{2}\right)^{t/t_{1/2}}$.

WORKED EXAMPLE 24.8

Scheduling a nuclear-medicine scan

Technetium-99m (half-life 6.0 h) is the most widely used medical radioisotope. A dose of 800 MBq is prepared at 08:00. What is its activity at 14:00 and at 20:00? What is the decay constant?

Solution

- 14:00 is 6.0 h later, one half-life: 400 MBq.
- 20:00 is 12 h later, two half-lives: 200 MBq.

$$\lambda = \frac{\ln 2}{6.0 \text{ h}} = 0.116 \text{ h}^{-1} = 3.2 \times 10^{-5} \text{ s}^{-1}.$$

Evaluate. Tc-99m is nearly ideal for imaging. Its 140 keV gamma rays escape the body easily and are detected efficiently by gamma cameras. Its 6-hour half-life is long enough for a scan, but short enough that the patient's dose stays low. It emits no alpha or beta particles. It is produced on site from a molybdenum-99 “generator”, which hospitals replace weekly.

WORKED EXAMPLE 24.9

Radiocarbon dating

Carbon-14 ($t_{1/2} = 5730 \text{ y}$) is produced continually in the atmosphere, and living organisms keep a constant ratio of ^{14}C to ^{12}C . After death, the ^{14}C decays and is not replaced. A wooden artefact has 25% of the ^{14}C activity of living wood. How old is it?

Solution

25% is $(\frac{1}{2})^2$, so two half-lives have passed: $t = 2(5730) = 11\,460$ years. In general,

$$t = \frac{t_{1/2}}{\ln 2} \ln \frac{A_0}{A}.$$

Radiation dose and safety

The **absorbed dose** is the energy deposited per kilogram of tissue, measured in grays (1 Gy = 1 J/kg). Different radiations do different amounts of biological damage for the same energy deposited: alpha particles do about 20 times more than X-rays, gamma rays or beta particles. The **equivalent dose**, in **sieverts (Sv)**, includes this weighting.

EXPOSURE	TYPICAL DOSE
Natural background (per year)	about 2–3 mSv
Chest X-ray	0.02–0.1 mSv
Transatlantic flight	about 0.05 mSv
CT scan of the abdomen	about 8–10 mSv
Annual limit for radiation workers	20 mSv
Radiotherapy, to the tumour only	50–70 Gy, in fractions
Acute whole-body dose, often fatal without treatment	about 4–5 Gy

The three rules of radiation protection are **time** (minimise it), **distance** (intensity falls off as $1/r^2$) and **shielding** (lead for X-rays and gamma rays, plastic for beta particles). Medical practice follows the ALARA principle: keep doses “as low as reasonably achievable”, consistent with getting the diagnostic information needed.

IN PRACTICE

Nuclear medicine and radiotherapy.

- **PET imaging.** Fluorine-18 ($t_{1/2} = 110$ min) attached to glucose (FDG) accumulates in metabolically active tissue, such as tumours, the active brain and inflamed tissue. Each positron it emits annihilates with an electron, producing two 511 keV photons travelling in opposite directions, since momentum must be conserved. A ring of detectors records them in coincidence and reconstructs where each annihilation happened.
- **Radioiodine therapy.** Iodine-131 (8 days, β^- and γ) is taken up by the thyroid, where its beta particles destroy overactive or cancerous thyroid tissue, while its gamma rays allow imaging.
- **External-beam radiotherapy.** Medical linear accelerators produce 6–20 MV X-ray beams shaped to the tumour. Proton beams (Chapter 18) deposit most of their dose in the Bragg peak. Dividing the treatment into daily fractions lets healthy tissue repair itself between doses.
- **Sterilisation.** Gamma rays from cobalt-60 sterilise syringes, implants and surgical supplies in their sealed packaging.

CHAPTER SUMMARY

- Photon: $E = hf = hc/\lambda$ ($hc = 1240$ eV nm) and $p = h/\lambda$. Ionising radiation starts above about 10 eV.
- Photoelectric effect: $K_{\max} = hf - \phi = eV_s$, with threshold $f_0 = \phi/h$.
- X-rays: $\lambda_{\min} = hc/eV$. Compton shift $\Delta\lambda = (h/m_e c)(1 - \cos\theta)$. Attenuation $I = I_0 e^{-\mu x}$, with half-value layer $\ln 2/\mu$.
- de Broglie: $\lambda = h/p$. Uncertainty: $\Delta x \Delta p \geq \hbar/2$.
- Bohr hydrogen: $E_n = -13.6/n^2$ eV and $r_n = n^2 a_0$. Photon energy = level difference.
- Nucleus: $R = 1.2A^{1/3}$ fm. Binding energy Δmc^2 , with $1 \text{ u} = 931.5$ MeV. B/A peaks near iron, so both fission and fusion release energy.
- Decay: $N = N_0 e^{-\lambda t}$, $t_{1/2} = \ln 2/\lambda$, activity $A = \lambda N$ in becquerels. Dose in Gy (J/kg). Equivalent dose in Sv.

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P24.1

Very intense red light shining on potassium ejects no electrons, yet weak blue light does. Explain why, using the photon model.

P24.2

Why don't we notice the wave nature of a thrown ball, when electrons show it clearly? Support your answer with an order-of-magnitude calculation.

P24.3

Why are X-rays and gamma rays called “ionising radiation”, while radio waves and microwaves are not, even at very high intensities?

P24.4

After two half-lives, is a radioactive sample “gone”? What fraction remains after ten half-lives? Why do hospitals often store short-lived radioactive waste for about 10 half-lives before disposing of it as ordinary waste?

LEVEL B – STANDARD PROBLEMS

P24.5

A 1.0 mW helium–neon laser emits light at 633 nm. How many photons does it emit each second?

P24.6

In a photoelectric experiment, light of 300 nm gives a stopping potential of 2.03 V, and light of 400 nm gives 1.00 V. Use these data to find Planck's constant and the work function of the metal.

P24.7

Thermal neutrons from a nuclear reactor move at about 2200 m/s. Find their de Broglie wavelength, and explain why they are useful for studying the structure of crystals and biological molecules.

P24.8

An electron microscope uses 200 keV electrons. Verify the wavelength found in Example 24.4. Then find the wavelength you would get, wrongly, by ignoring relativity.

P24.9

Find the wavelengths of the first three lines of hydrogen's Balmer series ($n = 3, 4, 5 \rightarrow 2$). Which colours are they?

P24.10

At 100 keV, lead has an attenuation coefficient of about 60 cm^{-1} . (a) Find the half-value layer. (b) What fraction of the beam passes through a 0.50 mm lead apron? (c) What thickness reduces the beam to 0.10%?

P24.11

PET scheduling. An FDG dose (fluorine-18, $t_{1/2} = 110 \text{ min}$) is calibrated at 09:00. The patient is injected at 10:30 and needs 370 MBq. (a) What activity must the dose have at 09:00? (b) Why are the two annihilation photons emitted in opposite directions, and why does each have an energy of 511 keV?

P24.12

A 0.50 g sample of carbon from an ancient bone has a ^{14}C activity of 0.050 Bq. Living carbon has an activity of 0.23 Bq per gram. Estimate the age of the bone.

LEVEL C – CHALLENGE PROBLEMS

P24.13

The Compton edge. (a) Find the largest possible kinetic energy that a 511 keV photon can give to an electron in a single Compton scattering, and the energy of the scattered photon in that case. (b) Explain why PET detectors must use dense, high- Z crystals rather than plastic scintillators.

P24.14

The Bohr model. (a) Using Coulomb's force as the centripetal force, together with the condition $m_e v r = n \hbar$, derive $r_n = n^2 a_0$ with $a_0 = 4\pi\epsilon_0 \hbar^2 / (m_e e^2)$. (b) Show that $E_n = -13.6 \text{ eV} / n^2$, and evaluate a_0 .

P24.15

The energy of fission. For the reaction $^{235}\text{U} + n \rightarrow ^{141}\text{Ba} + ^{92}\text{Kr} + 3n$, find the energy released. Use the masses ^{235}U 235.043930 u, n 1.008665 u, ^{141}Ba 140.914411 u and ^{92}Kr 91.926156 u. Then find the energy released per kilogram of ^{235}U fully fissioned, and compare it with burning coal (about $3 \times 10^7 \text{ J/kg}$).

P24.16

Fusion. (a) Find the energy released in the reaction $\text{D} + \text{T} \rightarrow ^4\text{He} + n$. Use the masses D 2.014102 u, T 3.016049 u, ^4He 4.002603 u and n 1.008665 u. (b) Estimate the Coulomb barrier that the nuclei must overcome, taking their centres to be 3.0 fm apart at contact. (c) Find the temperature at which kT equals this barrier. Explain why fusion actually happens at much lower temperatures, around 10^8 K .

P24.17

Effective half-life. A radiopharmaceutical leaves the body by biological excretion, with a biological half-life T_b , as well as by radioactive decay, with a physical half-life T_p . (a) Show that the effective half-life satisfies $\frac{1}{T_{\text{eff}}} = \frac{1}{T_p} + \frac{1}{T_b}$. (b) Evaluate it for iodine-131 in the thyroid ($T_p = 8.0 \text{ d}$, $T_b = 80 \text{ d}$) and for a Tc-99m tracer ($T_p = 6.0 \text{ h}$, $T_b = 24 \text{ h}$).

P24.18

The dose from a bone scan. A patient is injected with 740 MBq of Tc-99m. Assume that it stays in the body and decays completely, that each decay emits one 140 keV photon, and that half of this energy is absorbed in the body. (a) How many nuclei decay in total? (b) Estimate the whole-body absorbed dose for a 70 kg patient. (c) Compare it with the annual natural background dose. (Biological excretion, ignored here, would actually reduce the dose.)

All worked examples and practice problems in these notes are original.
Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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